

2.1 Definitions, Uses, and Benefits of Computer Networks

2.1.1 Definition of a Computer Network

A **computer network** is a collection of interconnected computing devices that can exchange data and share resources through communication links. The devices in a network may include desktop computers, laptops, mobile phones, servers, printers, CCTV systems, IP phones, ATMs, routers, switches, firewalls, and Internet of Things devices. Each participating device is usually called a **node**, and a node that sends or receives data for users is commonly called a **host**.

The mechanical function of a network begins when one device creates data and sends it over a transmission medium. The data may travel as electrical signals over copper cable, light pulses over fiber-optic cable, or electromagnetic waves through wireless media. Networking hardware forwards this data toward the destination. Networking software and protocols determine how the data is formatted, addressed, divided into packets, delivered, acknowledged, retransmitted if necessary, and finally presented to the correct application.

A network must satisfy certain communication expectations. It should deliver data to the correct destination, preserve accuracy, deliver within an acceptable time, and reduce delay variation for real-time applications such as voice and video. These principles match the broad fundamentals of data communication: correct delivery, accuracy, timeliness, and jitter control. In practice, a network is judged not only by whether it connects devices, but by how reliably, securely, and efficiently it supports organizational work.

2.1.2 Important Network Terms

Term	Meaning	Example
Node	Any device connected to a network.	Laptop, router, printer, CCTV camera.
Host	An end device that sends or receives user data.	User PC, server, mobile phone.
Link	The communication path between two devices.	Ethernet cable, fiber link, Wi-Fi channel.
Medium	The physical or wireless path used to carry signals.	Twisted-pair cable, fiber, radio waves.
Protocol	A set of rules controlling communication.	TCP, IP, Ethernet, DNS, HTTP.
Topology	The physical or logical arrangement of devices and links.	Star, mesh, bus, ring.
Bandwidth	The theoretical capacity of a communication channel.	1 Gbps Ethernet interface.
Throughput	The actual amount of useful data transferred per unit time.	650 Mbps achieved on a 1 Gbps link.
Latency	Delay experienced by data while travelling from source to destination.	20 ms round-trip delay to a server.
Jitter	Variation in packet delay, especially important for voice/video.	Uneven audio during a call.

2.1.3 Uses of Computer Networks

The uses of computer networks have expanded from simple file sharing to the operation of entire digital economies. In organizations, networks connect employees to shared applications, databases, printers, storage systems, email platforms, and business systems. In education, networks provide access to e-learning platforms, digital libraries, virtual laboratories, and online examinations. In banking, networks connect branches, ATMs, mobile banking systems, card platforms, core banking systems, payment gateways, and security monitoring platforms. In healthcare, they support patient records, diagnostic imaging, telemedicine, and medical equipment integration.

Networks also make distributed computing possible. Instead of every device storing everything locally, users can access centralized or cloud-hosted resources. A small phone can use a remote server for authentication, a browser can retrieve a web

application from another country, and an organization can back up data to a disaster recovery site. This distributed design improves service availability and enables scale, but it also introduces the need for addressing, routing, access control, monitoring, and security.

- **Resource sharing:** printers, files, applications, storage, Internet links, and processing power can be shared among many users.
- **Communication:** email, messaging, voice over IP, video meetings, and collaborative platforms depend on networks.
- **Centralized management:** administrators can apply policies, patches, backups, authentication, and monitoring from central systems.
- **Business continuity:** organizations can replicate data to remote sites and continue operations during local failures.
- **Remote access:** users can securely connect to office systems from branches, homes, or mobile locations.
- **Scalability:** new users, departments, and services can be added without redesigning the whole computing environment.

2.1.4 Benefits of Computer Networks

The first major benefit of networking is **efficiency**. Shared resources reduce duplication. For example, instead of installing a separate printer for every user, an office can deploy network printers. Instead of copying files using USB drives, users can use shared storage with access permissions.

The second benefit is **speed of communication**. Reports, forms, images, and application data can move almost instantly across a campus or across countries.

The third benefit is **reliability through redundancy**. Network design can include multiple switches, routers, links, and data centers so that one failure does not stop all services.

The fourth benefit is **centralized control**. Network administrators can enforce authentication, password policies, software updates, logging, and security monitoring.

The fifth benefit is **collaboration**. Modern work increasingly depends on simultaneous access to shared documents, databases, and communication tools.

At the same time, benefits bring responsibilities. A poorly designed network can spread malware, expose confidential data, or create a single point of failure. Therefore, good network design balances usability, performance, cost, manageability, and security.

Deep Dive: Network Performance Is Not Only Bandwidth

People often believe that a faster link automatically means a better network. Bandwidth is only one factor. A 1 Gbps link with high packet loss, excessive delay, or poor routing may perform worse than a lower-bandwidth but stable link. For interactive applications such as remote desktop, voice calls, and banking transactions, latency and jitter can be as important as bandwidth. Performance analysis therefore considers bandwidth, throughput, latency, jitter, packet loss, error rate, and device processing capacity.

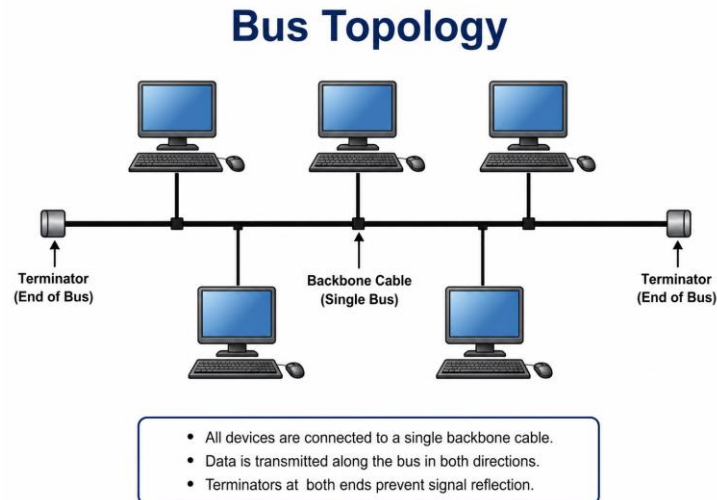
2.2 Network Topologies and Their Types

A **network topology** describes how network devices are arranged and how data flows between them. Topology can be discussed in two ways. **Physical topology** refers to the actual layout of cables, devices, and wireless links. **Logical topology** refers to the path that data follows, which may differ from the physical layout. For example, a modern Ethernet LAN often has a physical star layout because all devices connect to a central switch, but the logical forwarding decisions are made by the switch based on MAC addresses.

Topologies matter because they influence cost, performance, troubleshooting, scalability, and fault tolerance. A topology that is cheap for a small classroom may be unsuitable for a data center. A topology that is highly resilient may be too expensive for a small office. Network designers select topologies by balancing business requirements, technical constraints, budget, and future growth.

2.2.1 Bus Topology

In a **bus topology**, all devices share a single central communication cable called the **backbone**. Each device is attached to this backbone through a connector or tap. When a device transmits data, the signal travels along the shared medium and can be seen by all connected devices. Only the intended recipient accepts and processes the data; other devices ignore it. Traditional bus Ethernet used coaxial cable and terminators at both ends to prevent signal reflection.



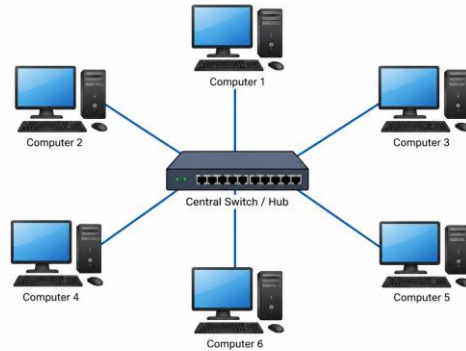
Bus topology is simple but sensitive to faults. Because all devices share the same medium, only one device should transmit at a time. If two devices transmit simultaneously, a collision may occur. Older Ethernet systems used collision detection mechanisms to manage this shared access. If the main backbone cable breaks, the entire network segment may fail. Troubleshooting can also be difficult because one physical fault can affect many users.

A real-world historical application of bus topology was early Ethernet using coaxial cable. It is rarely used for modern office LANs, but understanding bus topology is still useful because it explains shared media communication, collision domains, and why switched Ethernet became more practical.

2.2.2 Star Topology

In a **star topology**, each device connects to a central device such as a switch, hub, or wireless access point. Most modern LANs use some form of star topology. A user computer connects to an access switch, the switch connects to a distribution switch, and the distribution layer connects to routers, firewalls, or core switches. If one user cable fails, usually only that user is affected.

Star Topology



The central device receives frames or signals and forwards them. If the central device is a hub, it repeats incoming signals to all ports. If it is a switch, it learns MAC addresses and forwards frames only to the appropriate port whenever possible. This improves performance and reduces unnecessary traffic. The main weakness is that the central device can become a single point of failure unless redundancy is added.

Star topology is common in offices, banks, computer labs, call centers, and home networks. It is easy to manage because cables run to a central location such as a network rack. It is also easier to expand because new devices can be added by connecting them to available switch ports.

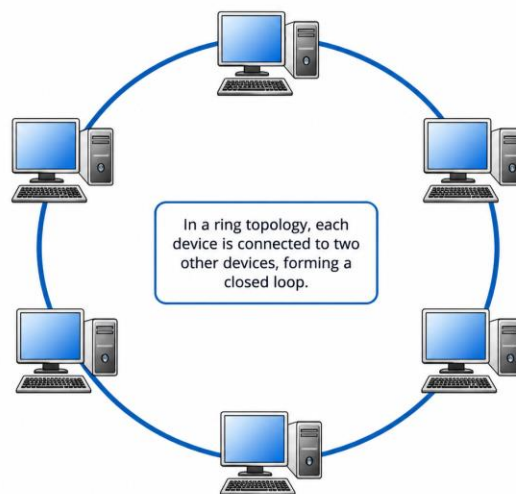
2.2.3 Ring Topology

In a **ring topology**, each device connects to two neighboring devices, forming a closed loop. Data travels around the ring from one device to the next until it reaches the destination. In some ring systems, a special control frame called a **token** is passed around the network. Only the device holding the token can transmit, which helps avoid collisions.

Ring topology can provide orderly access to the medium because devices take turns transmitting. However, a break in the ring can disrupt communication unless the system has fault-tolerant mechanisms such as dual rings or bypass circuits. Ring ideas have appeared in technologies such as Token Ring and some metropolitan or fiber network designs.

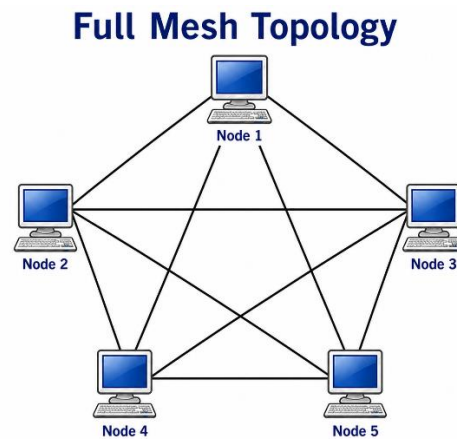
A useful application example is a campus or telecom fiber ring, where buildings are connected in a loop to provide path redundancy. If one segment fails, traffic may be redirected in the opposite direction if the ring supports protection switching.

Ring Topology



2.2.4 Mesh Topology

In a **mesh topology**, devices are interconnected with multiple paths. In a **full mesh**, every device has a direct link to every other device. In a **partial mesh**, only selected devices have multiple interconnections. Mesh topology is valued for reliability because if one path fails, another path may still be available.



Mesh networks require routing or switching logic to choose the best path. More links mean more redundancy, but also more cost and complexity. A full mesh of many devices becomes expensive because the number of links grows quickly. For n devices, a full mesh requires $n(n-1)/2$ links. This is why full mesh is usually limited to critical network cores or small groups of highly important devices, while partial mesh is used in larger networks.

Real-world applications include ISP backbones, data center fabrics, wireless mesh networks, and enterprise WANs. In banking, critical data center connectivity may use redundant paths between core routers, firewalls, and switches to avoid service interruption.

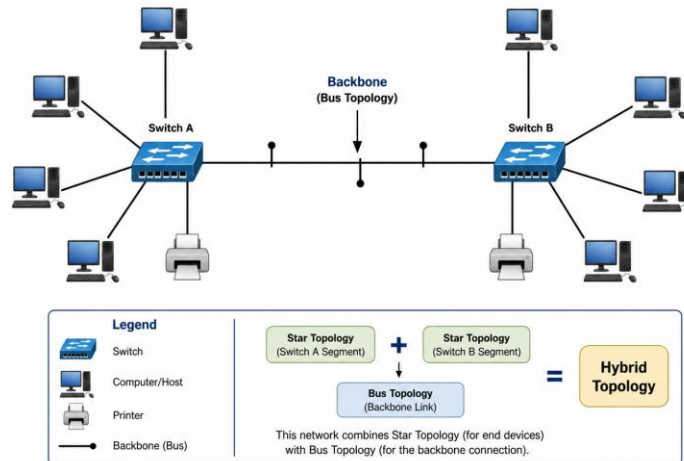
Deep Dive: Why Full Mesh Becomes Expensive

If there are 5 routers in a full mesh, 10 links are needed. If there are 10 routers, 45 links are needed. If there are 20 routers, 190 links are needed. Each link may require interfaces, cables, optics, configuration, monitoring, and troubleshooting. Therefore, practical networks often use partial mesh or hierarchical designs rather than connecting everything to everything.

2.2.5 Hybrid Topology

A **hybrid topology** combines two or more topology types. Most real networks are hybrid. A university campus may use star topology within each building, a fiber ring between buildings, and mesh connectivity at the data center core. A bank may use star topology inside branches, WAN links to head office, and redundant mesh between data centers.

Hybrid Topology



Hybrid topology exists because real requirements are mixed. A classroom needs simplicity and low cost. A server room needs high availability. A wide-area network needs provider connectivity and backup paths. A hybrid design allows each part of the network to use the topology that best fits its purpose.

The main challenge is management. Hybrid designs require documentation, addressing plans, monitoring, consistent standards, and skilled administrators. Without documentation, a hybrid network becomes difficult to troubleshoot because failures may pass through several layers and technologies.

Topology	Mechanical Function	Advantages	Limitations	Common Use
Bus	All devices share one backbone cable.	Low cable cost; simple for very small networks.	Backbone failure affects all; collisions; difficult troubleshooting.	Historical coaxial Ethernet; teaching shared media concepts.
Star	Each device connects to a central hub/switch/AP.	Easy to install, manage, and expand; single cable failure affects one node.	Central device failure affects connected nodes unless redundant.	Modern Ethernet LANs, computer labs, offices.
Ring	Devices form a loop; data passes node to node.	Orderly access; can support predictable behavior.	Break may disrupt ring unless protected.	Fiber rings, historical Token Ring, metro networks.
Mesh	Devices have multiple interconnections.	High redundancy and fault tolerance.	Costly; complex cabling and routing.	Data centers, ISP backbones, critical WANs.
Hybrid	Combination of multiple topology patterns.	Flexible and realistic for large organizations.	Requires careful design and documentation.	Campuses, banks, enterprises, hospitals.

Network Situation	Physical Topology	Logical Topology
Computers connected to one coaxial backbone cable	Bus	Bus
Computers connected to a hub	Star	Bus
Computers connected to a switch	Star	Star
Devices connected in a circular loop	Ring	Ring
Token Ring using central MAU	Star	Ring

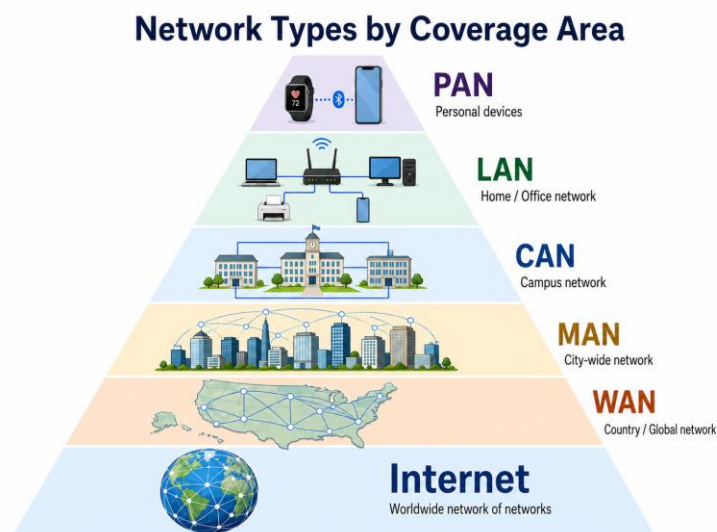
Network Situation	Physical Topology	Logical Topology
Routers connected with multiple redundant links	Mesh	Mesh
Office LANs connected to a campus backbone	Hybrid	Hybrid

Analogy Corner: Topology as Classroom Seating

A bus topology is like students sharing one long bench: if the bench breaks, everyone is affected. A star topology is like all students reporting to one teacher: communication is organized, but the teacher is central. A ring topology is like passing a notebook around a circle. A mesh topology is like every student having direct communication with many others. A hybrid topology is the real classroom, where different communication patterns are used for different activities.

2.3 Types of Computer Networks

Computer networks can be classified by the area they cover, the ownership model, the technology used, and the purpose they serve. The most common classification is based on geographical scope: **PAN**, **LAN**, **CAN**, **MAN**, and **WAN**. This classification helps understand design requirements. A network connecting a smartwatch to a phone has different constraints from a network connecting bank branches across a country.



2.3.1 Personal Area Network (PAN)

A **Personal Area Network (PAN)** connects devices around an individual, usually within a range of a few meters. Examples include a smartphone connected to wireless earbuds, a smartwatch synchronized with a phone, or a laptop connected to a mobile hotspot. Technologies used in PANs include Bluetooth, USB tethering, infrared in older systems, and short-range Wi-Fi connections.

PANs are designed for low power, short distance, and convenience. They typically prioritize ease of pairing and mobility rather than high throughput or enterprise-grade management. Security is still important because weak pairing or unauthorized access can expose personal data.

2.3.2 Local Area Network (LAN)

A **Local Area Network (LAN)** connects devices within a limited area such as a home, office, classroom, branch, or building. LANs are usually privately owned and managed by an individual or organization. Ethernet and Wi-Fi are the most common LAN technologies. A LAN may include switches, wireless access points, printers, file servers, and Internet gateways.

LAN communication often uses MAC addresses at the data link layer. Switches learn which MAC addresses are reachable through which ports and forward Ethernet frames accordingly. LANs generally provide high speed and low latency compared to wide-area links. In real organizations, LANs are segmented using VLANs to separate departments, users, servers, guests, and management traffic.

2.3.3 Campus Area Network (CAN)

A **Campus Area Network (CAN)** connects multiple LANs within a campus-like environment. A university, hospital complex, industrial estate, corporate head office, or bank head-office campus may use a CAN. It is larger than a LAN but usually smaller than a MAN. It is typically owned and managed by one organization.

A CAN often uses fiber-optic backbone links between buildings, distribution switches, and core switches. The design may include redundant paths so that failure of one cable or switch does not disconnect an entire building. CANs require structured cabling, network rooms, documentation, addressing plans, and centralized monitoring.

2.3.4 Metropolitan Area Network (MAN)

A **Metropolitan Area Network (MAN)** covers a city or metropolitan region. It may connect campuses, offices, government buildings, data centers, or service provider points of presence. MANs are often operated by telecom providers, ISPs, municipalities, or large organizations with city-wide infrastructure.

MANs commonly use fiber rings, metro Ethernet, microwave links, or provider-managed networks. The design must consider distance, provider service-level agreements, redundancy, and routing. A bank connecting multiple branches within Kathmandu Valley through a service provider metro Ethernet service is an example of MAN-like connectivity.

2.3.5 Wide Area Network (WAN)

A **Wide Area Network (WAN)** covers large geographical areas such as regions, countries, or continents. WANs connect branch offices, data centers, cloud platforms, disaster recovery sites, and remote users. The Internet is the largest example of global internetworking, but organizations also build private WANs using leased lines, MPLS, VPNs, SD-WAN, satellite links, or service provider networks.

WANs depend heavily on routing. Routers determine how packets move between networks. WAN links usually have higher latency and lower bandwidth than LAN links, so application design, compression, caching, QoS, and redundancy become important. WAN design also requires strong security because traffic may cross provider or public networks.

2.3.6 Other Useful Classifications

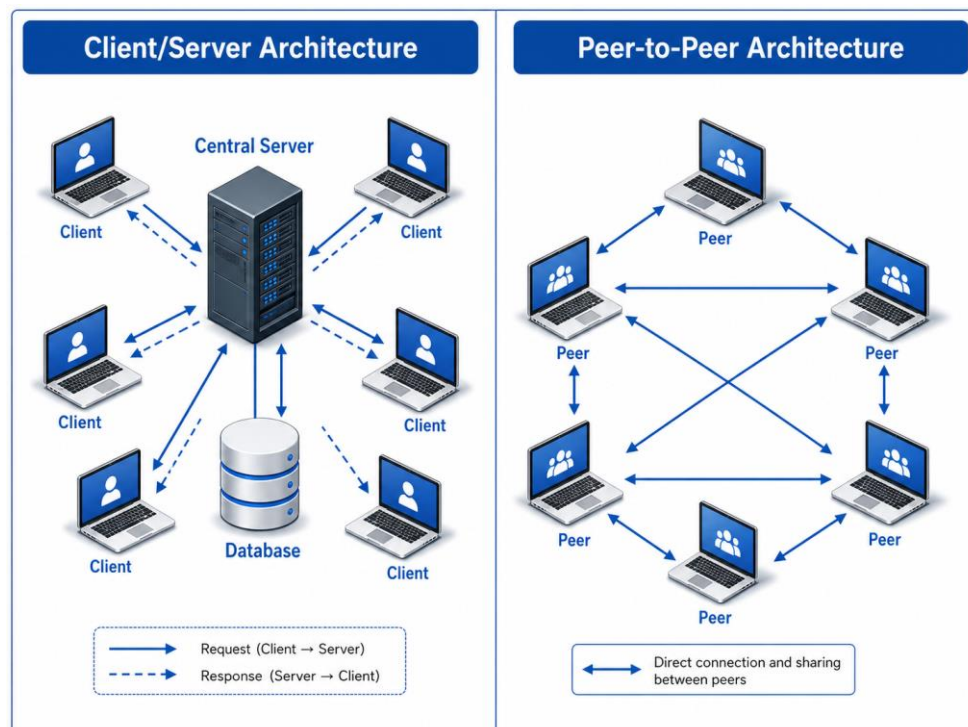
In addition to geographical categories, networks may be classified by ownership and function. An **intranet** is a private internal network using Internet technologies for organizational users. An **extranet** extends controlled access to selected external partners, vendors, or customers. A **VPN** creates an encrypted logical network over another network, commonly the Internet. A **SAN (Storage Area Network)** is a specialized high-speed network for storage traffic in data centers. A **WLAN** is a wireless LAN based on Wi-Fi technologies.

Network Type	Coverage Area	Typical Technologies	Ownership	Example
PAN	Few meters around a person	Bluetooth, USB, hotspot	Individual	Phone to smartwatch.
LAN	Room, floor, building, branch	Ethernet, Wi-Fi	Private organization/home	Office computers connected to switches.
CAN	Campus or group of buildings	Fiber backbone, Ethernet, Wi-Fi	Single organization	University buildings connected together.

MAN	City/metropolitan area	Metro Ethernet, fiber ring, microwave	Provider or large organization	City-wide branch connectivity.
WAN	Country, region, global	MPLS, leased line, VPN, SD-WAN, satellite	Provider/enterprise	Head office connected to branches nationwide.
SAN	Data center storage environment	Fibre Channel, iSCSI, NVMe-oF	Enterprise	Servers accessing shared storage arrays.

2.4 Networking Types: Client/Server and Peer-to-Peer

Networks can also be classified by how responsibilities are distributed among devices. Two important models are **client/server** and **peer-to-peer (P2P)**. This distinction is not about cable layout; it is about service architecture. It answers the question: who provides the service, who consumes it, and who controls access?



2.4.1 Client/Server Networking

In a **client/server network**, specialized computers called **servers** provide services, and other devices called **clients** request and consume those services. A server may provide authentication, file storage, email, database access, web applications, printing, DNS, DHCP, backup, or security monitoring. Clients may be desktop computers, laptops, mobile devices, or applications.

The client initiates a request and the server responds. For example, when a web browser opens a website, the browser acts as an HTTP client and the web server responds with web content. When a user logs into a domain, the workstation contacts an authentication server. When an application retrieves customer records, it sends queries to a database server. Client/server design centralizes data and control, making administration easier.

The main advantages are manageability, security, scalability, backup control, and consistent service. The main disadvantages are cost, dependency on server availability, and the need for skilled administration. In a bank, client/server design is essential because access control, audit logs, data integrity, and centralized management are critical.

2.4.2 Peer-to-Peer (P2P) Networking

In a **peer-to-peer network**, devices communicate directly and may act as both clients and servers. A user computer may share a folder, printer, or file directly with another user computer. There may be no dedicated central server. This model is simple and inexpensive for small environments, but it becomes difficult to manage as the number of devices grows.

Each peer controls its own shared resources and permissions. This means configuration is distributed. If one peer is turned off, resources stored on that peer become unavailable. Security is often weaker because access control may be inconsistent across devices. P2P is useful in small home networks, temporary collaboration, and some distributed applications, but it is not ideal for regulated enterprise environments where control and auditing are important.

Criteria	Client/Server	Peer-to-Peer
Control	Centralized; server controls services and access.	Distributed; each peer controls its own resources.
Cost	Higher due to servers, licenses, administration.	Lower for very small networks.
Security	Stronger centralized authentication, authorization, logging.	Inconsistent security across devices.
Scalability	Scales better with proper server infrastructure.	Becomes difficult as peers increase.
Reliability	Can be highly reliable with redundancy and backup.	Depends on individual peers being online.
Best Use	Banks, universities, enterprises, hospitals.	Home networks, small groups, temporary sharing.

Architecture	Modern Applications	How It Works
Client/Server Architecture	Online banking, mobile banking apps, e-commerce websites, cloud email, social media, Google Workspace, Microsoft 365, Netflix, YouTube, hospital systems	The client such as browser/mobile app sends a request to a central server , and the server processes the request and returns the response. This model follows a request-response approach.
Peer-to-Peer Architecture	BitTorrent, blockchain networks, cryptocurrency networks	Devices called peers communicate and share resources directly with each other instead of depending fully on one central server.

Analogy Corner: Restaurant Model

Client/server networking is like a restaurant: customers request food, and the kitchen prepares and serves it in an organized way. Peer-to-peer networking is like a picnic where everyone brings and shares food directly. The picnic is simple and friendly for a small group, but a large event needs a kitchen, staff, inventory, hygiene controls, and management.

2.5 Overview of Protocols and Standards

A **protocol** is a formal set of rules that governs communication between devices. It defines what messages look like, what order they are exchanged in, how errors are handled, how addresses are interpreted, and how both sides know that communication has succeeded or failed. Without protocols, devices may be physically connected but still unable to understand one another.

A **standard** is an agreed specification adopted by recognized bodies, vendors, or communities so that products from different manufacturers can interoperate. Protocols become useful at scale when they are standardized. For example, Ethernet standards allow network cards and switches from different vendors to communicate. IP standards allow routers across the world to forward packets. Wi-Fi standards allow phones, laptops, and access points from many vendors to connect.

The distinction is important. A protocol is the rule system; a standard is the accepted, documented, and widely implemented form of that rule system. Standards prevent vendor lock-in, reduce cost, improve compatibility, and support innovation because new products can build on existing agreements.

2.5.1 Elements of a Protocol

Protocols are commonly described through three elements: **syntax**, **semantics**, and **timing**. Syntax defines the structure or format of data, such as fields in a header. Semantics defines the meaning of each field and the action to be taken when a message is received. Timing defines when data can be sent and how fast it can be sent. Together, these elements make communication predictable.

Protocol Element	Letter Analogy	Meaning
Syntax	The format of the letter	Where to write the sender name, receiver name, date, subject, body, and signature
Semantics	The meaning of the content	What the subject means, what action is requested, and how the receiver should respond
Timing	The sending and response schedule	When the letter should be sent, how quickly it should be delivered, and when the reply is expected

For example, an IP packet header includes fields such as source address, destination address, time-to-live, and protocol number. The syntax defines where these fields are located. The semantics define that the destination address identifies the receiving network host and that time-to-live helps prevent endless looping. Timing and behavior rules determine how routers process and forward the packet.

2.5.2 Standards Organizations

Networking standards are developed and maintained by several organizations. The **ISO** is associated with the OSI model and many international standards. The **IEEE** develops important LAN and MAN standards, including Ethernet under IEEE 802.3 and Wi-Fi under IEEE 802.11. The **IETF** develops Internet standards through Request for Comments documents, commonly called RFCs. The **ITU-T** works on telecommunications standards. The **ANSI**, **TIA**, and **EIA** have influenced cabling, signaling, and telecommunications specifications.

Organization	Area of Contribution	Example
ISO	International standards and reference models.	OSI reference model.
IEEE	LAN/MAN standards and physical/data link technologies.	Ethernet 802.3, Wi-Fi 802.11.
IETF	Internet protocols and RFCs.	IP, TCP, UDP, DNS-related RFCs.
ITU-T	Telecommunication standards.	Signaling, optical transport, telephony standards.
TIA/EIA	Telecommunications cabling and infrastructure.	Structured cabling categories and wiring standards.

2.5.3 Common Protocols in Modern Networks

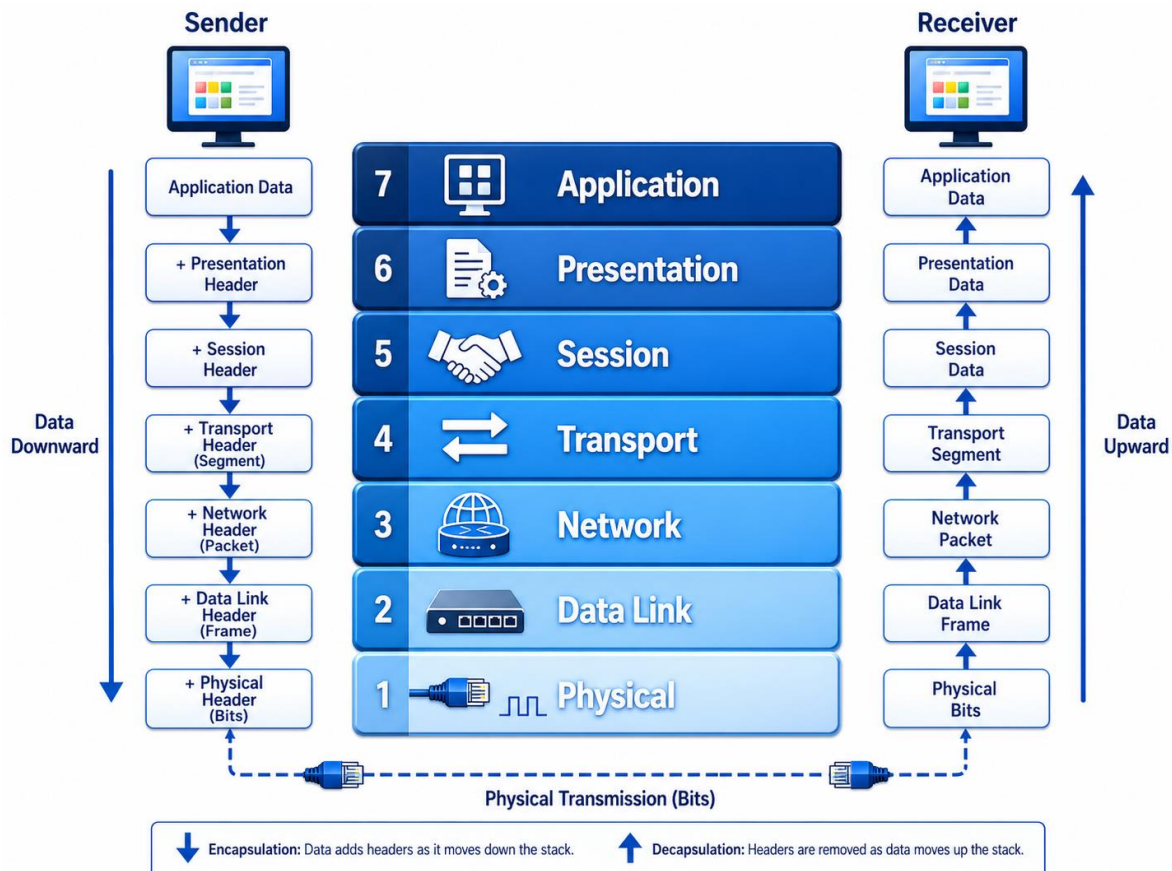
Protocol	Layer/Function	Main Purpose	Real-World Example
Ethernet	Data Link / LAN	Frames data on wired LANs using MAC addresses.	PC connected to an office switch.
IP	Network	Provides logical addressing and routing between networks.	Packet travelling from branch to data center.
TCP	Transport	Reliable, connection-oriented delivery with sequencing and acknowledgments.	Web login, email, file transfer.
UDP	Transport	Connectionless delivery with lower overhead.	DNS queries, voice/video streams.
HTTP/H TTPS	Application	Web communication between browser and server.	Opening an online banking page.
DNS	Application	Translates domain names to IP addresses.	Resolving example.com before opening a website.

DHCP	Application/Network service	Automatically assigns IP configuration to clients.	Laptop receives IP after joining Wi-Fi.
SMTP	Application	Transfers email between mail servers.	Sending email through organization mail gateway.
ARP	Network/Data Link boundary	Maps IPv4 addresses to MAC addresses in a LAN.	PC finds router MAC address before sending traffic.

2.6 OSI Reference Model

The **Open Systems Interconnection (OSI) reference model** is a seven-layer framework used to understand and design network communication. It was developed as a conceptual model for interoperability among different systems. The OSI model is not itself a single protocol used by the Internet; rather, it is a teaching and design model that divides communication into manageable layers. Each layer performs a specific set of functions and provides services to the layer above it.

The value of OSI is intellectual clarity. Network problems can be analyzed layer by layer. If a user cannot access a website, the issue may be physical connectivity, data link configuration, IP addressing, routing, transport-layer blocking, DNS resolution, TLS, or application failure. The OSI model gives a structured method for troubleshooting rather than guessing.



2.6.1 Layer 1: Physical Layer

The **Physical Layer** is responsible for transmitting raw bits over a communication medium. It defines electrical, mechanical, procedural, and functional characteristics of the interface between device and medium. It concerns voltage levels, signal timing, connectors, cable types, radio frequencies, modulation, bit rate, and physical topology.

This layer converts bits into signals. In copper cable, bits may be represented as electrical voltage changes. In fiber, bits may be represented as light pulses. In wireless systems, bits may be represented as modulated radio waves. If the physical layer fails, higher layers cannot communicate because no reliable signal path exists. A disconnected cable, damaged fiber, weak Wi-Fi signal, or failed network interface are physical-layer problems.

A real-world example is connecting a computer to a switch using a Cat6 Ethernet cable. Link lights, cable continuity, connector quality, and negotiated speed are all physical-layer concerns.

2.6.2 Layer 2: Data Link Layer

The **Data Link Layer** is responsible for node-to-node or hop-to-hop delivery over a single link. It organizes raw bits into **frames**, handles physical addressing through MAC addresses, detects some transmission errors, and controls access to the medium. Ethernet and Wi-Fi operate largely at this layer.

When an IP packet needs to travel across a LAN, the data link layer encapsulates it into an Ethernet frame. The frame includes a source MAC address, destination MAC address, type/length field, payload, and frame check sequence. Switches use MAC addresses to forward frames within the LAN. If a switch receives a frame, it checks its MAC address table and forwards the frame out the correct port, floods it if the destination is unknown, or filters it if the destination is on the same incoming port.

A real-world example is a laptop sending a frame to the default gateway. The laptop uses ARP to learn the router interface MAC address, then sends the Ethernet frame to that MAC address.

2.6.3 Layer 3: Network Layer

The **Network Layer** is responsible for source-to-destination delivery across multiple networks. It provides logical addressing and routing. The Internet Protocol (IP) is the dominant network-layer protocol. Routers operate primarily at this layer because they forward packets between different networks.

The network layer adds source and destination IP addresses to packets. Routers read the destination IP address, consult routing tables, decrement time-to-live values, and forward packets toward the next hop. Unlike a switch, which forwards frames inside a LAN, a router connects different networks and makes path decisions.

A real-world example is traffic from a branch office PC reaching a data center application. The packet may pass through a branch router, ISP network, firewall, core router, and server network before reaching the destination.

2.6.4 Layer 4: Transport Layer

The **Transport Layer** provides process-to-process communication between applications on different hosts. It uses port numbers to identify applications and may provide reliability, sequencing, flow control, and error recovery. TCP and UDP are the two most common transport-layer protocols.

TCP establishes a logical connection, numbers of data segments, acknowledges received data, retransmits lost segments, and controls the sending rate. UDP provides a simpler connectionless service without built-in reliability, making it useful for applications that prefer speed or handle reliability themselves. Port numbers allow multiple applications on one host to communicate at the same time. For example, TCP port 443 is commonly used for HTTPS, while UDP port 53 is commonly used for DNS queries.

A real-world example is logging into an online application. TCP ensures that the web data arrives in order and without missing pieces. For a live voice call, UDP may be preferred because late packets may be less useful than timely packets.

2.6.5 Layer 5: Session Layer

The **Session Layer** establishes, manages, and terminates communication sessions between applications. It is concerned with dialog control, synchronization, and recovery points.

Session functions help two systems maintain organized conversation. For example, a file transfer or database session may need to resume from a known point if communication is interrupted. In many modern systems, session management is implemented within application protocols, authentication systems, web tokens, cookies, or application frameworks.

2.6.6 Layer 6: Presentation Layer

The **Presentation Layer** is responsible for data representation, translation, encryption/decryption, and compression. It ensures that data sent by one system can be understood by another system. Different systems may use different character encodings, data formats, or byte orders, and the presentation layer addresses such differences.

presentation-layer functions may convert data formats, compress multimedia, or encrypt data before transmission. Examples include character encoding such as Unicode, image formats such as JPEG or PNG, data formats such as JSON or XML, and security transformations such as TLS encryption. In the TCP/IP world, many presentation functions are handled by libraries and application protocols.

2.6.7 Layer 7: Application Layer

The **Application Layer** provides network services directly to user applications. It does not mean the application program itself; rather, it refers to protocols and services that allow applications to use the network. Examples include HTTP/HTTPS for web access, DNS for name resolution, SMTP for email sending, FTP/SFTP for file transfer, and SNMP for network management.

the application layer creates messages meaningful to the user or application. For example, an HTTP client sends a request such as GET /index.html, and the web server returns a response. A DNS client asks for the IP address of a domain name, and a DNS server replies. Application-layer protocols are closest to user activity and business services.

OSI Layer	Main Function	Data Unit	Common Examples	Typical Device/Concern
7 Application	Network services for applications.	Message/Data	HTTP, DNS, SMTP, FTP, SNMP	Web server, DNS server.
6 Presentation	Translation, encryption, compression.	Data	TLS, JPEG, JSON, Unicode	Encryption, format conversion.
5 Session	Session establishment and management.	Data	RPC, session tokens, checkpoints	Login/session control.
4 Transport	End-to-end process delivery.	Segment/Datagram	TCP, UDP	Ports, reliability, flow control.
3 Network	Logical addressing and routing.	Packet	IP, ICMP, IPsec	Router, routing table.
2 Data Link	Framing and local delivery.	Frame	Ethernet, Wi-Fi, PPP	Switch, MAC address.
1 Physical	Transmission of raw bits.	Bits	Copper, fiber, radio	Cable, connector, signal.

Deep Dive: Encapsulation and Decapsulation

When data is sent, each lower layer adds its own control information. Application data becomes a transport segment, then a network packet, then a data-link frame, then physical bits. This is encapsulation. At the receiver, each layer removes and interprets the corresponding control information. This is decapsulation. Encapsulation allows each layer to focus on its own responsibility without needing to know every detail of other layers.

The 7 layers of the OSI model

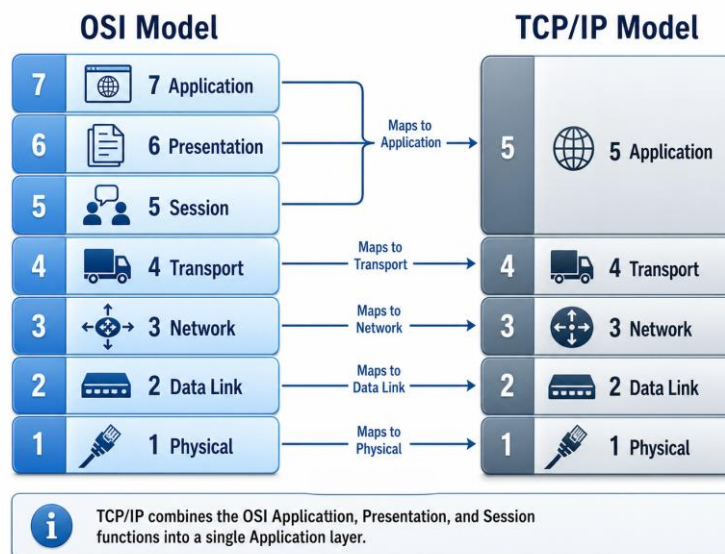
Layer	Description	Example protocols	Message Name
7 Application Layer	Network process to application.	HTTP, FTP, SSH, Telnet, DNS, etc.	Data
6 Presentation Layer	Recognizing data.	HTML, MPEG, JPEG, IMAP, TLS, etc.	
5 Session Layer	Session establishment and termination.	Sockets, Winsock, APIs, etc.	
4 Transport Layer	End-to-end connectivity and reliability.	TCP, UDP	Segment (TCP) Datagram (UDP)
3 Network Layer	Logical addressing (IP) and path determination.	IPv4, IPv6, ICMP, IPsec, IGMP, OSPF, etc.	Packet
2 Data Link Layer	Physical addressing (MAC).	802.3 Ethernet, 802.11 Wireless, PPP, HDLC, etc.	Frame
1 Physical Layer	Media, signals and binary transmission.	100Base-FX, 1000Base-LX, RS-232, DSL, etc.	Bits

Figure 8. The 7 layers of the OSI model.

2.7 TCP/IP Model and Comparison with OSI

The **TCP/IP model** describes the protocol architecture used by the Internet and most modern networks. Unlike OSI, which is mainly a reference model, TCP/IP is implemented globally. It developed through practical use, research, and Internet deployment. TCP/IP is commonly described as a four-layer model or a five-layer model. For teaching, the five-layer view is often clearer because it separates physical and data link functions.

The five-layer TCP/IP view includes **Physical**, **Data Link**, **Network**, **Transport**, and **Application** layers. The TCP/IP application layer combines the functions that OSI separates into session, presentation, and application layers. This does not mean those functions disappeared; rather, they are implemented inside application protocols, libraries, and software systems.



2.7.1 TCP/IP Layers

The **Physical Layer** in TCP/IP represents the transmission of bits through physical or wireless media. The **Data Link Layer** handles local delivery using frames and hardware addressing. The **Network Layer**, mainly IP, provides logical addressing and routing across interconnected networks. The **Transport Layer** provides process-to-process communication using TCP or UDP. The **Application Layer** contains user-facing network protocols such as HTTP, DNS, SMTP, DHCP, FTP, SSH, and SNMP.

When a user opens a website, the application layer creates an HTTP or HTTPS request. The transport layer uses TCP to establish reliable delivery to the web server port. The network layer uses IP addresses to route the packet. The data link layer uses local MAC addresses to deliver frames to the next hop. The physical layer transmits the bits through cable, fiber, or wireless signals.

2.7.2 Why TCP/IP Became Dominant

TCP/IP became dominant because it was implemented, tested, and scaled through the growth of the Internet. It supported internetworking among diverse networks and allowed independent networks to connect using a common protocol layer: IP. While OSI remains a powerful educational and design framework, TCP/IP is the practical architecture used in everyday networking.

TCP/IP also supports modularity. Different link-layer technologies can carry IP packets, including Ethernet, Wi-Fi, fiber links, cellular networks, and satellite systems. Different applications can use TCP or UDP depending on their requirements. This flexibility is a major reason the Internet can connect many technologies into one global system.

OSI Model	TCP/IP Model	Comparison
7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.	Usually 4 or 5 layers; five-layer teaching view uses Physical, Data Link, Network, Transport, Application.	OSI is more detailed conceptually; TCP/IP is more practical and widely implemented.
Developed as a reference model for open systems.	Developed from practical Internet protocols.	OSI helps learning and troubleshooting; TCP/IP runs real networks.
Separates Session and Presentation layers.	Combines those functions into Application layer.	Modern applications handle sessions, formats, and encryption internally or through libraries.
Strict layer descriptions.	More flexible protocol suite.	TCP/IP protocols can be mixed according to system needs.
Used heavily for education and structured troubleshooting.	Used for Internet, enterprise, cloud, and most LAN/WAN communication.	Both models are important for network professionals.

Analogy Corner: OSI as a Detailed Building Plan, TCP/IP as the Working City

OSI is like a detailed architectural drawing that labels every floor and function clearly. TCP/IP is like the actual city where people live, roads operate, and services run. The plan helps us understand structure, but the city is where real traffic moves every day.

2.8 Networking Hardware: NIC, Hub, Repeater, Switch, Bridge, and Router

Networking hardware provides the physical and logical movement of data. Some devices operate mainly at the physical layer, some at the data link layer, and some at the network layer. Understanding these devices helps students troubleshoot networks and design appropriate architectures. A common mistake is to treat all network boxes as the same. In reality, a hub, switch, and router make very different forwarding decisions.

[IMAGE PLACEHOLDER: Networking Devices by OSI Layer]

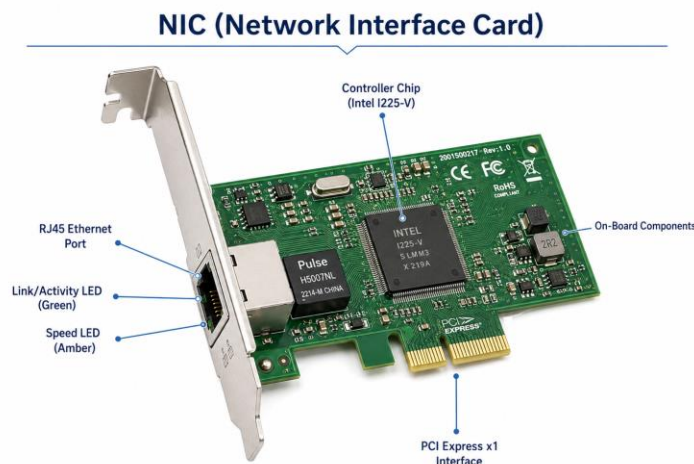
Image Generation Prompt:

Create a professional 3D isometric diagram mapping networking devices to OSI layers: NIC at Layer 1/2, hub and repeater at Layer 1, bridge and switch at Layer 2, router at Layer 3. Include small device icons and arrows showing data flow. Use clean academic colors and labels.

2.8.1 Network Interface Card (NIC)

A **Network Interface Card (NIC)** is the hardware component that connects a device to a network. It may be a physical Ethernet adapter, a wireless adapter, or an integrated chip on the motherboard. Every NIC has a hardware address called a **MAC address**, used for local network communication.

The NIC converts data from the computer into signals suitable for the medium and converts received signals back into data. An Ethernet NIC transmits and receives frames over cable. A wireless NIC transmits and receives radio signals through an antenna. NICs operate mainly at the physical and data link layers. Their settings may include speed, duplex, VLAN tagging, driver configuration, and MAC address information.



A real-world example is a laptop Ethernet adapter connecting to an office switch at 1 Gbps full duplex. If the NIC driver fails, the cable is unplugged, or duplex negotiation is incorrect, network performance or connectivity may be affected.

2.8.2 Hub

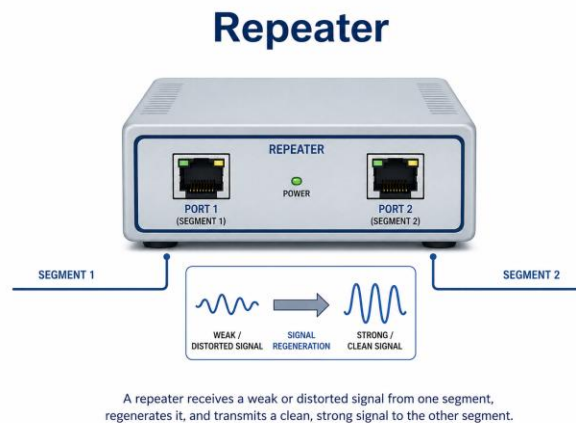
A **hub** is a simple physical-layer device that connects multiple Ethernet devices but does not intelligently forward frames. When a hub receives a signal on one port, it repeats that signal out all other ports. It does not learn MAC addresses and does not reduce collisions.



A hub creates a shared collision domain. All connected devices compete for the same bandwidth. Because a hub sends traffic to every port, it is inefficient and insecure compared with a switch. Hubs are mostly obsolete in modern networks, but they are useful historically for understanding shared Ethernet and collision behavior.

2.8.3 Repeater

A **repeater** is a physical-layer device that regenerates or amplifies signals to extend the distance of a network link. Signals weaken as they travel through media due to attenuation. A repeater receives a weak signal, cleans or regenerates it, and sends it onward.

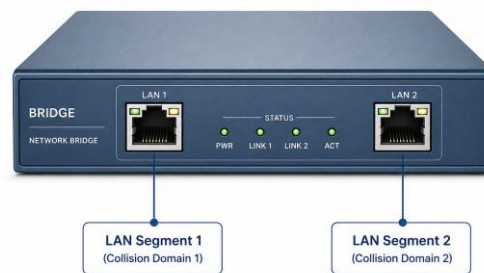


Repeaters do not interpret frames, MAC addresses, or IP addresses. They work with signal levels and timing. In modern networks, repeater-like functions are built into switches, wireless extenders, media converters, and optical transport equipment. A real-world example is a wireless repeater used to extend Wi-Fi coverage in a building, although such extension may reduce performance if not designed carefully.

2.8.4 Bridge

A **bridge** is a data-link-layer device used to connect and filter traffic between network segments. It reads MAC addresses and decides whether to forward or filter frames. Traditional bridges had fewer ports than modern switches, but the logic is similar: learn MAC addresses and reduce unnecessary traffic.

Bridge



A bridge builds a table that maps MAC addresses to ports or segments. If a frame is destined for a device on the same segment, the bridge can filter it. If it is destined for another segment, the bridge forwards it. Bridges helped improve early LAN performance by dividing collision domains.

2.8.5 Switch

A **switch** is a multiport data-link-layer device that connects devices within a LAN and forwards frames based on MAC addresses. Modern switches are central to Ethernet LANs. They learn MAC addresses by examining the source address of incoming frames and recording the port on which each address appears. When a frame arrives, the switch checks the destination MAC address and forwards the frame only to the appropriate port if known.

Switch



Switches reduce collisions, improve bandwidth usage, and support full-duplex communication. Managed switches can also support VLANs, trunking, spanning tree, port security, link aggregation, quality of service, and monitoring. Some advanced switches, called Layer 3 switches, can also perform routing functions.

A real-world example is an access switch in a branch office connecting user PCs, printers, IP phones, and wireless access points. The switch may separate voice and data traffic using VLANs and forward uplink traffic to a router or firewall.

2.8.6 Router

A **router** is a network-layer device that connects different networks and forwards packets based on IP addresses. Routers maintain routing tables that identify where packets should be sent next. Unlike switches, which primarily use MAC addresses within a LAN, routers use logical addresses to move traffic between networks.

Router



When a router receives a packet, it examines the destination IP address, chooses the best route, decrements the time-to-live value, updates link-layer information for the next hop, and forwards the packet through the selected interface. Routers can connect LANs to WANs, branches to head office, and private networks to the Internet. Many routers also perform NAT, access control, VPN termination, and quality-of-service functions.

A real-world example is a branch router connecting a bank branch LAN to the head office or data center through service provider links. If routing is misconfigured, users may have local connectivity but fail to reach remote applications.

Device	Main OSI Layer	Forwarding Basis	Key Function	Modern Status
NIC	Layer 1/2	Interface and MAC address	Connects host to network and converts data/signals.	Essential in every networked device.
Hub	Layer 1	No addressing; repeats signals.	Connects devices in shared medium.	Mostly obsolete.
Repeater	Layer 1	Signal regeneration.	Extends signal distance.	Used in specific wired/wireless/optical contexts.
Bridge	Layer 2	MAC address table.	Connects LAN segments and filters frames.	Concept integrated into switches.
Switch	Layer 2 mostly	MAC address table.	Connects LAN devices and forwards frames efficiently.	Core device in modern LANs.
Router	Layer 3	IP routing table.	Connects different networks and forwards packets.	Essential for WAN/Internet/inter-VLAN routing.

Deep Dive: Collision Domain vs Broadcast Domain

A hub does not separate collision domains; all connected devices share the same collision space. A switch separates collision domains because each port can operate independently. However, a switch does not automatically separate broadcast domains unless VLANs are used. A router separates broadcast domains because it forwards packets between networks and normally does not forward Layer 2 broadcasts.

2.9 Basic Networking Commands

Networking commands allow administrators and students to observe and troubleshoot network behavior. These commands are practical tools that connect theory with reality. A student may learn about IP addressing in class, but commands such as ipconfig, ping, traceroute, arp, and nslookup reveal how addressing, routing, and name resolution work on a real device.

Command syntax differs slightly between Windows, Linux, and macOS. However, the diagnostic logic is similar. First, check local configuration. Second, test local connectivity. Third, test gateway connectivity. Fourth, test remote IP connectivity. Fifth, test DNS name resolution. Sixth, examine routes, ARP entries, ports, and active connections.

2.9.1 ipconfig, ifconfig, and ip addr

ipconfig on Windows and **ifconfig** or **ip addr** on Linux/macOS display network interface configuration. They show IP address, subnet mask or prefix length, default gateway, DNS information, MAC address, and interface status depending on the option used. This is usually the first command to run when troubleshooting because a device without a valid IP address cannot communicate properly beyond local link-level behavior.

Windows:

```
ipconfig
ipconfig /all
ipconfig /release
ipconfig /renew
```

Linux:

```
ip addr show
ip route show
ifconfig    # older systems, may require net-tools
```

A real-world example is a laptop that cannot access the Internet. If `ipconfig` shows an address beginning with 169.254.x.x, the device likely failed to obtain an IP address from DHCP and assigned itself an automatic private address.

2.9.2 ping

ping tests reachability using ICMP Echo Request and Echo Reply messages. It helps determine whether a destination is reachable and provides round-trip time. Ping is useful but not absolute: some firewalls block ICMP even when the service is available.

Windows/Linux/macOS:

```
ping 127.0.0.1
ping 192.168.1.1
ping 8.8.8.8
ping example.com
```

A structured use of ping begins with the loopback address 127.0.0.1 to test the local TCP/IP stack. Then ping the device own IP address, then the default gateway, then a remote IP address, and finally a domain name. If remote IP works but domain name fails, DNS may be the issue.

2.9.3 tracert and traceroute

tracert on Windows and **traceroute** on Linux/macOS show the path packets take to a destination by listing intermediate routers. These commands use TTL/hop-limit behavior to identify each hop. They help locate where delay or failure occurs along a path.

Windows:

```
tracert example.com
```

Linux/macOS:

```
traceroute example.com
tracepath example.com
```


A real-world example is a branch user reporting that a central application is slow. Traceroute may show whether delay begins inside the branch, at the ISP, near the data center, or at the destination network.

2.9.4 nslookup and dig

nslookup and **dig** query DNS servers. DNS translates human-readable domain names into IP addresses. If a user can ping an IP address but cannot open a website by name, DNS resolution should be tested.

```
Windows/Linux/macOS:  
nslookup example.com  
nslookup example.com 8.8.8.8
```

```
Linux/macOS:  
dig example.com  
dig example.com A  
dig example.com MX
```

A real-world example is email delivery troubleshooting. MX records identify mail servers for a domain. DNS errors can prevent users from reaching websites, sending email, or accessing internal applications by name.

2.9.5 arp

arp displays and manages the Address Resolution Protocol cache. ARP maps IPv4 addresses to MAC addresses on a local network. When a device wants to send an IP packet to another device on the same LAN, or to the default gateway, it needs the destination MAC address for the Ethernet frame.

```
Windows:  
arp -a  
arp -d *  
  
Linux:  
ip neigh show  
ip neigh flush all
```

A real-world example is duplicate IP troubleshooting. If two devices use the same IP address, ARP entries may change unexpectedly, causing intermittent connectivity. Security teams also monitor abnormal ARP behavior because ARP spoofing can support man-in-the-middle attacks on local networks.

2.9.6 netstat and ss

netstat and **ss** display active network connections, listening ports, routing tables, and protocol statistics. They are useful for identifying whether a service is listening, which remote systems are connected, and what ports are in use.

```
Windows:  
netstat -ano  
netstat -rn  
  
Linux:  
ss -tulnp  
ss -ant  
ip route show
```

A real-world example is a web server that is unreachable. The administrator can check whether the server is listening on TCP port 80 or 443. If the service is not listening, the issue is local to the server or application. If it is listening, firewall or routing checks may be required.

2.9.7 route and ip route

route on Windows and **ip route** on Linux display or modify routing tables. A routing table tells the host where to send packets for different destination networks. Most end-user devices have a default route pointing to the default gateway.

```
Windows:
route print

Linux:
ip route show
ip route get 8.8.8.8
```

A real-world example is a computer connected to both Wi-Fi and Ethernet. Incorrect route priority may send traffic through the wrong interface. Route inspection helps identify such problems.

Command	Main Use	Example Diagnosis
ipconfig /all or ip addr	View interface IP, gateway, DNS, MAC.	No IP address, wrong subnet, DHCP issue.
ping	Test reachability and round-trip time.	Gateway unreachable, remote host unreachable.
tracert/traceroute	Show path and hops to destination.	Find where WAN delay or failure occurs.
nslookup/dig	Test DNS resolution.	Name resolution failure.
arp -a / ip neigh	View IP-to-MAC mappings.	Duplicate IP, ARP spoofing suspicion.
netstat/ss	View listening ports and connections.	Application not listening on expected port.
route print / ip route	View routing table.	Wrong default gateway or route priority.

Deep Dive: A Simple Troubleshooting Sequence

A disciplined sequence prevents random guessing: (1) check link light or Wi-Fi association, (2) check IP configuration, (3) ping loopback, (4) ping own IP, (5) ping default gateway, (6) ping remote IP, (7) test DNS, (8) trace the route, (9) inspect ARP and routing tables, and (10) check application ports and firewall rules. This sequence loosely follows the OSI model from lower layers to higher layers.

Chapter Summary

- A **computer network** connects devices so that they can exchange data and share resources.
- Networks provide resource sharing, communication, centralized management, remote access, redundancy, collaboration, and scalability.
- **Topology** describes how devices and links are arranged physically or logically. Common types include bus, star, ring, mesh, and hybrid.
- Networks may be classified by scope: **PAN**, **LAN**, **CAN**, **MAN**, **WAN**, and specialized forms such as SAN and WLAN.
- **Client/server** architecture centralizes services, control, and security; **peer-to-peer** architecture distributes sharing among peers.
- **Protocols** define communication rules; **standards** make those rules interoperable across vendors and systems.

- The **OSI model** divides communication into seven conceptual layers and is useful for learning, design, and troubleshooting.
- The **TCP/IP model** is the practical protocol architecture of the Internet and most modern networks.
- **NICs, hubs, repeaters, bridges, switches, and routers** operate at different layers and perform different forwarding functions.
- Basic commands such as **ipconfig, ping, traceroute, nslookup, arp, netstat/ss, and route/ip route** help diagnose connectivity issues.

Review Questions

1. Define a computer network and explain why modern organizations depend on networking.
2. Differentiate between bandwidth, throughput, latency, and jitter.
3. Compare physical topology and logical topology with examples.
4. Explain the mechanical operation of bus topology. Why is it rarely used in modern LANs?
5. Why is star topology widely used in Ethernet LANs?
6. Differentiate between full mesh and partial mesh topology.
7. Classify PAN, LAN, CAN, MAN, and WAN using coverage area and real-world examples.
8. Compare client/server and peer-to-peer network architectures.
9. Define protocol and standard. Why are standards important in networking?
10. List the seven OSI layers and describe the main function of each layer.
11. Explain encapsulation and decapsulation using the OSI or TCP/IP model.
12. Compare OSI and TCP/IP models. Why is TCP/IP considered more practical?
13. Differentiate between a hub, switch, and router.
14. What is the role of a NIC in network communication?
15. Describe a step-by-step command sequence to troubleshoot a user who cannot access a website.

Practical Activities

- **Activity 1: Draw Topologies.** Draw bus, star, ring, mesh, and hybrid topologies for a small organization. Identify likely single points of failure.
- **Activity 2: Identify Network Scope.** Classify examples such as Bluetooth earbuds, office Ethernet, university backbone, city fiber ring, and national branch network as PAN, LAN, CAN, MAN, or WAN.
- **Activity 3: Packet Tracer Lab.** Create a small star topology with PCs, a switch, and a router. Assign IP addresses and test connectivity using ping.
- **Activity 4: Command-Line Diagnosis.** On a Windows or Linux system, record IP address, gateway, DNS server, ARP cache, route table, and active connections.

- **Activity 5: OSI Troubleshooting Case.** For the problem “user cannot access internal web application,” identify possible causes at each OSI layer.

Mini Case Study: Designing a Small Office Network

A small office has 35 employees, one file server, one printer, Wi-Fi users, and an Internet connection. A suitable design would use a star topology with one or more switches at the center. User PCs, printers, and access points would connect to the switch. A router or firewall would connect the LAN to the Internet. The file server would use a static IP address, while user devices could obtain addresses through DHCP. DNS would be configured for name resolution. VLANs could separate staff, guest Wi-Fi, and server traffic if the switch supports VLANs.

Troubleshooting in this network would follow a layered method. If one user cannot connect, check cable, Wi-Fi, NIC, IP address, gateway, DNS, and application access. If all users cannot connect to the Internet, check the router/firewall, ISP link, default route, and DNS. This case shows how topology, network type, architecture, devices, models, and commands work together in real design.

Glossary for the Professional

ARP: Address Resolution Protocol; maps IPv4 addresses to MAC addresses on a local network.

Bandwidth: The theoretical maximum capacity of a communication link.

Bridge: A Layer 2 device that connects LAN segments and forwards or filters frames using MAC addresses.

Broadcast Domain: A network area where Layer 2 broadcast frames are received by all devices unless separated by routers or VLANs.

Bus Topology: A topology where all devices share a single backbone medium.

Client: A device or application that requests services from a server.

Client/Server Architecture: A model where centralized servers provide services to clients.

Collision Domain: A network segment where packet collisions can occur, especially in shared Ethernet media.

Data Link Layer: OSI Layer 2; responsible for framing, MAC addressing, and local delivery.

Default Gateway: The router address used by a host to reach destinations outside its local network.

DNS: Domain Name System; resolves domain names into IP addresses.

Encapsulation: The process of adding layer-specific headers or trailers as data moves down the protocol stack.

Ethernet: A widely used LAN technology standardized by IEEE 802.3.

Frame: The protocol data unit of the data link layer.

Hub: A Layer 1 device that repeats signals to all ports; mostly obsolete.

IP Address: A logical address used to identify devices at the network layer.

Jitter: Variation in packet delay, important for real-time communication.

LAN: Local Area Network; a network covering a small area such as an office or building.

MAC Address: Hardware address used at the data link layer.

Mesh Topology: A topology with multiple interconnections and redundant paths.

Network Layer: OSI Layer 3; responsible for logical addressing and routing.

NIC: Network Interface Card; connects a device to a network.

OSI Model: Seven-layer conceptual model for understanding network communication.

Packet: The protocol data unit of the network layer.

PAN: Personal Area Network; short-range network around a person.

Peer-to-Peer: A distributed model where devices share resources directly without a dedicated central server.

Protocol: A set of rules governing communication between systems.

Repeater: A Layer 1 device that regenerates signals to extend distance.

Router: A Layer 3 device that forwards packets between networks using IP addresses.

Server: A system or application that provides services to clients.

Star Topology: A topology where devices connect to a central device such as a switch.

Switch: A multiport Layer 2 device that forwards frames based on MAC address tables.

TCP/IP Model: The practical protocol architecture used by the Internet and modern networks.

Topology: The physical or logical arrangement of network devices and links.

WAN: Wide Area Network; a network covering large geographical areas.

Suggested References

- Behrouz A. Forouzan, Data Communications and Networking, Fourth Edition, McGraw-Hill Higher Education, 2007.
- Course syllabus and unit objectives provided for Data Communication and Networks.
- Relevant standards and documentation from ISO, IEEE, IETF RFCs, ITU-T, and TIA/EIA may be consulted for deeper study.